Tier-D Ranks 11–20: Two General Repairs and a 9/10 Closure

Tactix engineering handoff

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Objective and outcome

This round predeclared the next ten eligible Tier-D instances on which the frozen Kissat baseline outperformed Tactix, excluding the already closed ranks 1–10. The goal was to locate the earliest execution divergence with tracing, make general source-backed repairs, and preserve N1–N4 correctness and the fixed Tier B plus selected Tier C iteration guard.

The accepted release binary has SHA-256 `d39cc80fd3412dc407040a22a0c1553d8c14a0c10e21b8ffc1bb737f1ff7d399`. It improves this cohort from 6/10 to 9/10 checked solves under the pre-Phase-4.5 contract (16 GiB, 1,200-second solver timeout). PAR-2 falls from 10,576.236 to 4,140.349 seconds. All six SAT answers have checked models, all three UNSAT answers have checked proofs, and the campaign has zero certificate residue. Rank 19, `knight_18`, is the sole remaining timeout.

Before and after

The Kissat column is the frozen 32 GiB/5,000-second baseline projected to the debug contract: an instance counts only when its recorded time is at most 1,200 seconds and peak RSS is at most 16 GiB. All ten listed baselines are eligible. The Tactix before and after values are directly measured at 16 GiB/1,200 seconds. Times are solver wall seconds.

Rank	Result	Short name	Kissat	Tactix before	Tactix after
11	UNSAT	hwmcc20-mul2	0.266	2.923	0.417
12	UNSAT	hwmcc17-xits	5.574	TIMEOUT	13.644
13	SAT	128_100	68.965	84.725	87.689
14	SAT	64_200	69.168	79.560	86.531
15	SAT	mdp-32	81.163	2.421	2.470
16	SAT	noL-11-2	124.551	TIMEOUT	74.158
17	SAT	three-cubes-42	134.112	TIMEOUT	924.856
18	UNSAT	linked-list-swap	149.905	393.182	118.244
19	SAT	knight_18	153.363	TIMEOUT	TIMEOUT
20	SAT	cabp-715	189.311	413.425	432.340

The initial evidence is `benchmarks/runs/20260817-1350-tierd-next10-predeclared/`; the accepted cumulative evidence is `benchmarks/runs/20260817-1630-tierd-next10-header-final/`. The unusually wide rank-17 and rank-18 timing variation across repeated runs is real cluster cotenancy noise, so the table supports coverage and bounded-runtime claims, not a fine-grained speedup claim for those two rows.

Repair 1: Congruence closure cost on rank 12

Tracing and stage timing placed rank 12's gap inside Congruence rather than the search policy. Three general mismatches against Kissat composed into the timeout.

1. Tactix repeatedly answered binary membership by scanning mixed watch lists. Its `binary_ids` table could not be reused because that table is a size-monotone proof-identity archive whose entries deliberately survive clause deletion. Congruence now constructs a scoped exact live-pair index once and kernel binary add/delete hooks maintain it for the application. Binary scans also stop at the first large watch when the watch-list prefix is canonical.

2. A new root unit reopened every extracted gate. Kissat indexes gates by their literals and revisits only affected gates. Tactix now builds an immutable input/output occurrence schedule and requeues gates mentioning either polarity of each newly assigned literal. Every gate remains initially queued, so the selective path cannot omit first inspection.

3. Tactix ran speculative propagation before every structurally derived Congruence lemma even in proofless and DRAT modes. Kissat emits these lemmas directly. Tactix now does the same for proofless/DRAT execution; native LRAT retains replay because it needs ordered antecedent identifiers. The complete DRAT proof, not the structural argument alone, remains the correctness oracle.

The focused proof-bearing witness is `benchmarks/runs/20260817-1505-tierd-rank12-direct-drat/`: rank 12 is UNSAT in 15.148 seconds in that campaign, with its 59 MB DRAT proof accepted. The cumulative accepted-pin rerun is 13.644 seconds. New unit tests cover mixed watch extraction, canonical binary-prefix termination, and a late XOR unit revisiting an earlier dependent AND gate.

Repair 2: sparse DIMACS headers on rank 16

The first macro-state divergence was present before search: Kissat had 1,086 active variables while Tactix had 1,416. The 330 additional Tactix variables were declared by the DIMACS header but never occurred in any clause. Kissat does not import them; Tactix reserved them and admitted them to the decision population, perturbing every later search clock with propagation-free choices.

At end of input, a never-imported header variable is now marked eliminated and removed from DefaultPolicy's active population. Its stable external identity and total output-model width are preserved. This needs no reconstruction witness because the variable occurs nowhere, and total model extension may assign either value. The focused checked witness is `benchmarks/runs/20260817-1625-tierd-rank16-header-only-checked/`: SAT in 71.680 seconds with all 7,823 original clauses checked over the 1,419-variable model. The accepted cumulative rerun is 74.158 seconds. A parser regression test holds both the reduced active population and preserved external identity.

Divergence findings and disciplined negative result

Rank 17's preprocessing states are nearly identical, and the two solvers make the same first 47,048 decisions. The first observed choice divergence is at decision 47,049 after conflict 1,229: Kissat chooses external literal `-1035500`, while Tactix chooses `-1050035`. The surrounding seven choices are identical. Inspection of Kissat's `bump.c`, `decide.c`, and VMTF queue operations suggested that

Tactix might advance an assigned search cursor differently. An isolated implementation of Kissat's exact old-neighbour move-to-front rule reproduced the same choice divergence at the same decision. That hypothesis was therefore falsified; the experiment was reverted and never promoted. The cumulative accepted candidate nevertheless solves the instance, but the remaining queue/bump-state mismatch is not claimed as causally repaired.

For rank 19, all conflict clauses match through conflict 225 and all decisions match through decision 68,706. After the common positive decision on variable 102931, the first 38 traced implications match. The 39th binary-watch visit is the first ordered mismatch. Later in that same propagation, the common binary `(-102556,-101136)` and the common eight-literal clause `(101136,101269,101545,100663,101559,101464,100514,100613)` become opposing unit implications. Kissat reaches the large clause first, saves `+101136`, and conflicts on the binary; Tactix reaches the binary first, saves `-101136`, and conflicts on the large clause. The residual saved phase causes the first decision mismatch at decision 68,719: Kissat chooses `+101136`, Tactix `-101136`. This is evidence of a post-preprocessing binary-watch emission-order mismatch, not a branching admission or phase-policy defect. The existing `ORDER-WATCH-1` parity disposition is therefore reopened at its documented trigger.

Correctness and regression evidence

The accepted iteration guard is `benchmarks/runs/20260817-1635-BC-next10-header-final/`. Tier B remains 28/28 (14 SAT, 14 UNSAT) with aggregate solve time 1,089.550 seconds. The predeclared mechanism-relevant Tier C subset remains 5/5 (3 SAT, 2 UNSAT) with aggregate solve time 1,353.205 seconds. Every result is proof/model checked and both sets have zero certificate residue. Runtime totals vary from earlier co-scheduled repetitions, but there is no coverage or correctness regression.

The final local release build reproduces the accepted binary hash. CTest passes 72/72; two checker-dependent proof tests are reported as skipped because their optional external checkers are absent. Documentation size, citation, section reference, progress, and diff-whitespace checks pass. The source changes are limited to Congruence/kernel support, sparse-header finalization, focused tests, and the implementation/status documentation.

Limitations and unresolved risks

- This is an iteration gate, not the canonical 32 GiB/5,000-second Tier-D gate. Baseline memory eligibility is simulated from peak RSS and does not identify the exact time at which a process would cross 16 GiB.
- Rank 19 remains unsolved. Its first ordered implication, conflict, and decision divergences are localized, but the Congruence/Sweep/Substitute transaction that emitted the differently ordered binary suffix is not yet identified. Rank 17's first differing choice is likewise localized but not explained by a verified queue-state cause.
- Rank 17 closes at 924.856 seconds, leaving limited margin under the 1,200-second debug timeout. It should be treated as a coverage closure, not a robust performance win.
- The fixed five-instance Tier C set is a mechanism-relevant iteration guard, not a replacement for full Tier C or the phase gates. N4 ledger reconciliation and the canonical T4 comparisons remain open.

Next concrete actions

1. Trace rank 19's first post-preprocessing binary-watch-order mismatch back to the Congruence, Sweep, or Substitute transaction that emitted it. Compare that lifecycle with Kissat's dense save/extract/resume order.

2. Repair the responsible lifecycle generally, then rerun the focused witness, 28B+5C guard, and cumulative twenty Tier-D targets. Do not sort or tune specifically to this instance.

3. Reconcile the scheduling-parity ledger into the N4 verdict, then resume the remaining pre-Phase-4.5 Tier-D convergence work.